

Some types or combination of functions are similar, so we can expect to take their derivatives in a similar way. The main motivation for this is to save time and avoid tedious limit calculations. We will discuss the rules one at a time, but at the end they will all be listed out.

Constant functions

These are functions of the type $f(x) = C$ for some constant C .

We then have

$$f'(x) = \lim_{h \rightarrow 0} \frac{\overset{f(x+h)}{C} - \overset{f(x)}{C}}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

So $\frac{d}{dx}(C) = 0$ "derivative of a constant is 0"

This is expected as the derivative gives slope at any point and constant functions are straight lines with no slope.

Constant Rule: $\frac{d}{dx}(C) = 0$

Power functions

We know $\frac{d}{dx}(x) = 1$, $\frac{d}{dx}(x^2) = 2x$, $\frac{d}{dx}(x^3) = 3x^2$

One can show $\frac{d}{dx}(x^4) = 4x^3$ / try it yourself!

We start to see a pattern here, in fact the power rule is

Power Rule: $\frac{d}{dx}(x^n) = nx^{n-1}$

currently, we will look at the case where n is a natural number, but this rule works for all real numbers.

Proof: $f(x) = x^n$, $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h}$ "n choose k" $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

$$= \lim_{h \rightarrow 0} \frac{\cancel{x^n} + \binom{n}{1}x^{n-1}h + \binom{n}{2}x^{n-2}h^2 + \binom{n}{3}x^{n-3}h^3 + \dots + h^n - \cancel{x^n}}{h}$$
$$= \lim_{h \rightarrow 0} \frac{\binom{n}{1}x^{n-1}\cancel{h} + \binom{n}{2}x^{n-2}h^2 + \dots + h^{n-1}}{\cancel{h}} = \lim_{h \rightarrow 0} \binom{n}{1}x^{n-1} + \binom{n}{2}x^{n-2}h + \dots$$

$\xrightarrow{h=0} \rightarrow$

$$= \binom{n}{1}x^{n-1} = nx^{n-1}$$

Ex: $\frac{d}{dx}(x^{1234}) = 1234 x^{1233}$

Constant Multiple Rule

Let c be some number and f a differentiable function

$$\text{Let } g(x) = cf(x)$$

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h}$$

$$= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = cf'(x)$$

So,

$$\begin{aligned} \text{Constant multiple Rule: } \frac{d}{dx}(cf) &= c \frac{d}{dx}(f) \\ (cf)'(x) &= cf'(x) \end{aligned}$$

$$\underline{\text{Ex:}} \quad \frac{d}{dx}(3x^4) = 3 \frac{d}{dx}(x^4) = 3 \cdot (4x^3) = 12x^3$$

$$\frac{d}{dx}(-x) = - \frac{d}{dx}(x) = -1$$

Sum/Difference Rule

Consider $(f \pm g)(x)$ where f and g are differentiable

$$\begin{aligned}(f \pm g)'(x) &= \lim_{h \rightarrow 0} \frac{(f \pm g)(x+h) - (f \pm g)(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x+h) \pm g(x+h) - f(x) \mp g(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(f(x+h) - f(x)) \pm (g(x+h) - g(x))}{h} \\&= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \pm \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\&= f'(x) \pm g'(x)\end{aligned}$$

Sum & Difference Rule: $\frac{d}{dx}(f \pm g) = \frac{d}{dx}(f) \pm \frac{d}{dx}(g)$
 $(f \pm g)' = f' \pm g'$

With all the above rules, we can differentiate any Polynomial.

Ex: $f(x) = x^8 + 12x^5 - 4x^4 + 10x^3 - 6x + 5$

$$f'(x) = 8x^7 + 12(5x^4) - 4(4x^3) + 10(3x^2) - 6(1) + 0$$

$$f'(x) = 8x^7 + 60x^4 - 16x^3 + 30x^2 - 6$$

Ex: Find where tangent line is horizontal for the function $f(x) = x^4 - 6x^2 + 4$.

- Horizontal tangent is where $f'(x) = 0$ ^{Slope is 0}

So,

$$f'(x) = 4x^3 - 12x = 4x(x^2 - 3)$$

$$f'(x) = 0 \text{ if } \underline{x=0, \text{ or } x = \pm\sqrt{3}}$$

We can find the explicit points as $(0, 4)$, $(-\sqrt{3}, -5)$
 $(\sqrt{3}, -5)$

Ex: The position of a particle is given by $s(t) = 2t^3 - 5t^2 + 3t + 4$ Centimeters after t -seconds. Find acceleration after 2 seconds.

- Acceleration is the derivative of Velocity, which is the derivative of position.

$$v(t) = s'(t) = 6t^2 - 10t + 3$$

$$a(t) = v'(t) = s''(t) = 12t - 10$$

$$a(2) = 12(2) - 10 = \underline{\underline{14 \text{ cm/s}^2}}$$

Product Rule:

Let f and g be differentiable functions

$$(fg)'(x) = \lim_{h \rightarrow 0} \frac{(fg)(x+h) - (fg)(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - \cancel{f(x+h)g(x)} + \cancel{f(x+h)g(x)} - f(x)g(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x+h)g(x)}{h} + \frac{f(x+h)g(x) - f(x)g(x)}{h}$$

$$= \lim_{h \rightarrow 0} f(x+h) \frac{g(x+h) - g(x)}{h} + g(x) \frac{f(x+h) - f(x)}{h}$$

$$= f(x) g'(x) + g(x) f'(x)$$

$$= f'(x) g(x) + f(x) g'(x)$$

Product Rule: $(fg)' = f'g + fg'$

Ex: $f(x) = (6x^3)(7x^4) \longrightarrow f(x) = 42x^7$

$f'(x) = (18x^2)(7x^4) + (28x^3)(6x^3)$		$f'(x) = 294x^6$
$= 168x^6 + 126x^6$		
<u>$= 294x^6$</u>		

Quotient Rule

Let f and g be differentiable.

$$\left(\frac{f}{g}\right)'(x) = \lim_{h \rightarrow 0} \frac{\left(\frac{f}{g}\right)(x+h) - \left(\frac{f}{g}\right)(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h)}{h g(x+h)g(x)}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - \cancel{f(x)g(x)} + \cancel{f(x)g(x)} - f(x)g(x+h)}{h g(x+h)g(x)}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x)}{h g(x+h)g(x)} + \frac{f(x)g(x) - f(x)g(x+h)}{h g(x+h)g(x)}$$

$$= \lim_{h \rightarrow 0} \left[g(x) \frac{f(x+h) - f(x)}{h} - f(x) \frac{g(x+h) - g(x)}{h} \right] \frac{1}{g(x+h)g(x)}$$

$$= \left[g(x)f'(x) - f(x)g'(x) \right] \frac{1}{[g(x)]^2}$$

$$= \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$$

Quotient Rule: $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$

Ex:

$$h(x) = \frac{x^2 + x - 2}{x^3 + 6}$$

$$f(x) = x^2 + x - 2, \quad g(x) = x^3 + 6$$

$$f'(x) = 2x + 1, \quad g'(x) = 3x^2$$

$$h'(x) = \frac{(2x+1)(x^3+6) - (x^2+x-2)(3x^2)}{(x^3+6)^2}$$

Ex: $f(x) = \frac{1}{x^n}$

$$f'(x) = \frac{0 \cdot x^n - 1 \cdot n x^{n-1}}{x^{2n}} = \frac{-n x^{n-1}}{x^{2n}} = \frac{-n}{x^{n+1}} = -n x^{-n-1}$$

expected from
Power rule

Power Rule: $\frac{d}{dx}(x^n) = n x^{n-1}$ for any real number n .

Ex: $\frac{d}{dx}(x^\pi) = \pi x^{\pi-1}, \quad \frac{d}{dx}(\sqrt{x}) = \frac{d}{dx}(x^{1/2}) = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$

Ex: Find equation of tangent and Normal line to $f(x) = \frac{\sqrt{x}}{1+x^2}$ at the point $(1, \frac{1}{2})$

We need slope at $x=1$, that is $f'(1)$

$$f'(x) = \frac{(\frac{1}{2}x^{-1/2})(1+x^2) - \sqrt{x}(2x)}{(1+x^2)^2}$$

$$f'(1) = \frac{\frac{1}{2}(1^{-1/2})(1+1^2) - \sqrt{1}(2(1))}{(1+1^2)^2} = \frac{1-2}{2^2} = -\frac{1}{4}$$

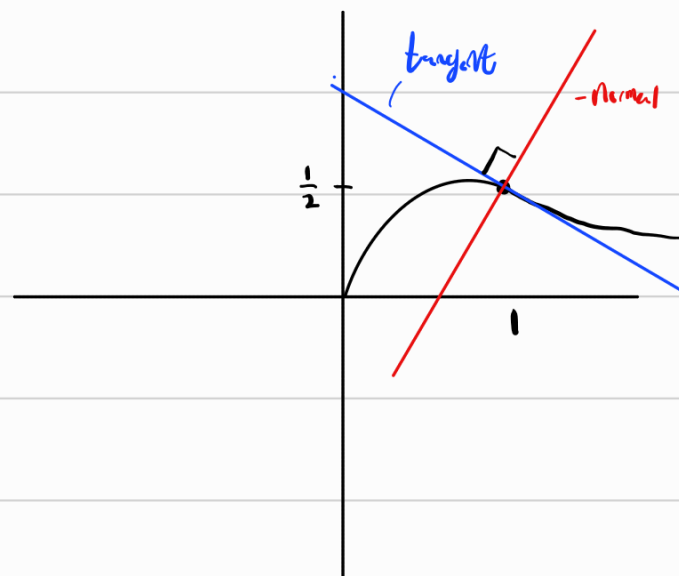
Tangent has slope $-\frac{1}{4}$, Normal has slope 4 ↖ $-\frac{1}{-\frac{1}{4}}$

Tangent: $y - \frac{1}{2} = -\frac{1}{4}(x-1)$

$$\boxed{y = -\frac{1}{4}x + \frac{3}{4}}$$

Normal: $y - \frac{1}{2} = 4(x-1)$

$$\boxed{y = 4x - \frac{7}{2}}$$



Ex: Where on the hyperbola $xy=12$ is the tangent line parallel to $3x+y=0$?

$3x+y=0 \rightarrow y=-3x$ so when does tangent have a slope of -3 ?

$$xy=12 \rightarrow y = \frac{12}{x} = 12x^{-1}$$
$$y' = -12x^{-2} = -\frac{12}{x^2}$$

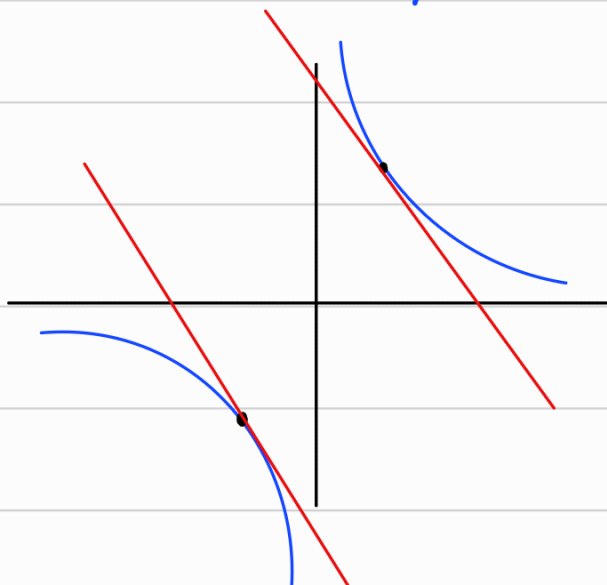
$$-\frac{12}{x^2} = -3$$

$$4 = x^2, \quad \underline{\underline{x = \pm 2}}$$

if $x=2, y=6$

$x=-2, y=-6$

So at $(2,6)$ and $(-2,-6)$ the tangent is parallel to $3x+y=0$



Derivative Rules:

Constant Rule

$$\frac{d}{dx}(c) = 0$$

Constant Multiple Rule

$$(cf)' = cf'$$

Power Rule

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Sum/Difference Rule

$$(f \pm g)' = f' \pm g'$$

Product Rule

$$(fg)' = f'g + fg'$$

Quotient Rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$